

Rahman Khandakar

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EDUCATION

The University of Texas at Austin

July 2025

Bachelor of Science, Electrical and Computer Engineering

GPA: 3.81/ 4.00

Projects

Sign Language Translation Wearable – Senior Design Projects

- Collaborated in a team of five to develop sensor-equipped gloves that captures ASL hand motions and translates them into English characters using machine learning winning the Capstone Design Award
- Created datasets by creating a Python program that labels glove sensor data into CSV files to train RNN models
- Designed and trained three PyTorch RNN models to translate sensor values into English text all with a 98% translation accuracy
- Implemented multithreaded Python integration code to process sensors values, enabling dynamic RNN models switching between left, right, or dual-hand translation modes based on glove positioning

Embedded Space Invaders Game

- Implemented player interaction using an LCD display for visuals, two buttons for input, an analog slide potentiometer via ADC for movement, and a resistor-based DAC speaker for sound effects
- Developed gameplay mechanics, including shooting logic, enemy animation, scoring, and bilingual language support, using C, ARM assembly, and interrupts on an TM4C microcontroller

Timer/stopwatch on FPGA board

- Designed and implemented a stopwatch/timer system on a Xilinx FPGA using Verilog and finite state machines
- Constructed clock divider modules using Verilog to provide precise timekeeping
- Developed 16-bit ripple carry adder and subtractor modules in Verilog as arithmetic building blocks ensuring accurate ALU operations
- Simulated and debugged functionality in Vivado using testbenches verifying correct timing, control logic, and output behavior across all states

Yash Shell

- Built a custom Unix-style shell in C with support for job control, signal handling, piping, and I/O redirection

Weather App

- Developed an Android weather forecast app in Java that retrieves hourly data from the Open-Meteo API and displays daily temperature, humidity, and rainfall summaries
- Designed networking logic with Volley to retrieve geolocated weather data based on user input allowing flexible weather forecasting for any user-specified location
- Created a clean and readable Android UI using TextViews, EditText, a button, enabling users to view daily weather summaries in a structured and visually appealing format

SKILLS

Programming Languages: C, C++, Java, Python, Verilog, Arm Assembly

Tools: Git